

10-COLOR FLOW CYTOMETRIC IMMUNOPHENOTYPING (ACUTE MYELOID LEUKEMIA)

Patient Name:	Gurmeet Singh	Date of Exam:	September 14, 2026
MRN / Patient ID:	RGCI-BMT-2026-7712	Age / Gender:	52 Yrs / Male
Blood Group & Rh:	B+ (Rh Positive)	Referring Doctor:	Dr. Dinesh Bhurani, MD, DM (Hemato-Oncology)
Clinical Diagnosis:	Suspected Acute Myeloid Leukemia (Diagnostic Workup)	Specimen / Source:	Bone Marrow Aspirate (Sodium Heparin)

Marker Panel	Expression %	Gating Intensity	Cellular Lineage	Status
CD45 vs SSC Gating	68.4 % Blasts	Dim Expression	Marrow Mononuclear	Leukemic Blast Gate
CD34 Progenitor	92.1 %	Bright Positive	Hematopoietic Stem	Primitive Blasts
CD117 (c-kit)	86.4 %	Moderate Positive	Myeloid Progenitor	Myeloid Lineage Positive
CD33 Myeloid	94.8 %	Bright Positive	Pan-Myeloid	Myeloid Lineage Positive
HLA-DR	88.2 %	Bright Positive	MHC Class II	Non-M3 Phenotype
CD19 / CD3 / CD7	< 2.0 %	Negative	Lymphoid Lineages	Negative for Lineage Infidelity

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- Immunophenotyping demonstrates 68.4% blast population expressing CD34, CD117, CD33, and HLA-DR.
- Flow profile confirms Acute Myeloid Leukemia with primitive progenitor immunophenotype (AML without maturation).
- Zero lymphoid cross-lineage infidelity detected on comprehensive 10-color antibody panel.
- Recommendation: Correlate with cytogenetics and initiate 7+3 Cytarabine + Daunorubicin induction.

Dr. Ritu Gupta, MD (Pathology)
Medical Technologist / Pathologist

Dr. Narendra Agrawal, MD, DM (Hemato-Oncology)
Consultant Hematopathologist

Dr. Dinesh Bhurani, MD, DM (Director of BMT)
Director of Cellular Therapy & BMT